Active recall workbook

CV-7 Conduction System and the Electrocardiogram

Cardiovascular System (CARD) · NPLEX Part I · NPLEX-P1-CARD-07

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-07-01-C01				
NPLEX-P1-CARD-07-01-C02				
NPLEX-P1-CARD-07-01-C03				
NPLEX-P1-CARD-07-01-C04				
NPLEX-P1-CARD-07-01-C05				
NPLEX-P1-CARD-07-01-C06				
NPLEX-P1-CARD-07-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C01-BD1

Draw the conduction pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C01 Activity: NPLEX-P1-CARD-07-01-C01-BD1 Date: _____

Without looking at your notes, draw the conduction pathway from the SA node to the ventricular myocardium. Name every structure in order and write beside each one where it sits in the heart, its conduction velocity, its intrinsic rate, and which ion carries its upstroke. Mark the one place where conduction slows, and write two sentences on why the cells there conduct slowly. Then mark the structure that insulates the atria from the ventricles and write why that makes the AV node and His bundle the only normal electrical route into the ventricles.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C01-BD2

What the AV delay buys and what happens without it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C01 Activity: NPLEX-P1-CARD-07-01-C01-BD2 Date: _____

The AV delay is about 0.13 s. Write what the ventricle gains during that delay and how much of resting filling it accounts for, then name the situations in which that contribution matters more than at rest. Next, the atria begin firing 400 times per minute. Write what the AV node does with those impulses and which property of nodal tissue lets it do that, then write what the ventricular rate and rhythm look like. Finish by writing what changes when an accessory pathway carries the same atrial impulses instead of the node.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C02-BD1

Draw the SA nodal action potential

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C02 Activity: NPLEX-P1-CARD-07-01-C02-BD1 Date: _____

Draw the SA nodal action potential with voltage on the vertical axis and time on the horizontal axis. Mark the maximum diastolic potential and the voltage at which the upstroke begins. Label phase 4, phase 0, and phase 3, and write beside each phase every current that moves during it and the direction each current carries charge. Write why nodal cells have no stable resting potential and which channel working myocytes have that nodal cells lack. Then write the two features this tracing does not have that a ventricular myocyte action potential does.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C02-BD2

Vagal and sympathetic drive at the node

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C02 Activity: NPLEX-P1-CARD-07-01-C02-BD2 Date: _____

List the three features of the pacemaker action potential that set firing rate. For acetylcholine at M2 receptors and for norepinephrine at beta-1 receptors, write which of those three features each one changes, in which direction, and through which second messenger and channel. Then write what each does to the PR interval and why. Finish with two patients: one receives a drug that blocks cardiac L-type Ca2+ channels, the other a drug that blocks HCN channels only. Write what happens to sinus rate, AV conduction, and ventricular contractility in each, and explain the difference from the currents involved.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-07-01-C02-RT01

Rebuild the sympathetic acceleration sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C02 Activity: NPLEX-P1-CARD-07-01-C02-RT01 Date: _____

Rebuild the six-step sequence that runs from norepinephrine arriving at the SA node to a faster heart rate, one step per line and in order. Beside the cAMP step, write which channel cAMP binds directly and what that binding does to the voltage at which the channel opens. Beside the protein kinase A step, write which channel it phosphorylates and what that does to the upstroke. Then draw the phase 4 slope before and after on one set of axes.

Rebuild from memory: Sympathetic acceleration of the SA node

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C03-BD1

Blood supply of the conduction system

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C03 Activity: NPLEX-P1-CARD-07-01-C03-BD1 Date: _____

Write out the blood supply of the conduction system from memory, one line per structure: SA node, AV node, bundle of His, right bundle branch, left anterior fascicle, left posterior fascicle. For each line name the artery or arteries and, where the notes gave one, the approximate percentage of hearts. Then write which two structures have more than one source and why that protects them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C03-BD2

Why inferior and anterior infarcts block differently

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C03 Activity: NPLEX-P1-CARD-07-01-C03-BD2 Date: _____

Two patients develop complete heart block. One has ST elevation in II, III, and aVF; the other has ST elevation in V1 to V4. For each patient write the occluded artery, the conduction tissue lost, the level of the block, the escape rhythm with its rate and QRS width, and whether the block is likely to recover. Then write the three mechanisms other than cell death that produce the bradycardia in the inferior infarct, and name the reflex that arises from the inferoposterior wall. Finish with the one anatomical variant that lets a circumflex occlusion cause AV nodal block.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C04-BD1

Draw and label one normal beat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C04 Activity: NPLEX-P1-CARD-07-01-C04-BD1 Date: _____

Draw one normal ECG beat and label the P wave, PR interval, PR segment, QRS complex, ST segment, T wave, QT interval, and U wave. Beside each label write the electrical event it records and its normal value or appearance. Write why atrial repolarization and the activity of the AV node and His bundle do not appear as waves. Then write why the ST segment is isoelectric and why the T wave points the same way as the QRS. Finish with the paper rules: the duration of a small box and a large box at 25 mm/s, and the voltage of 10 mm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C04-BD2

Work a strip from the numbers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C04 Activity: NPLEX-P1-CARD-07-01-C04-BD2 Date: _____

A tracing at 25 mm/s shows an upright P wave in leads I and II before every QRS. R waves are 3 large boxes apart. The PR interval is 6 small boxes, the QRS is 3 small boxes, and the QT is 8 small boxes. Lead I is positive, lead aVF is negative, and lead II is positive. Calculate the rate, the PR, the QRS, and the QT in seconds, show your working, and correct the QT for rate using the Bazett formula. State whether each value is normal. Then place the mean QRS axis within a range of degrees using the three limb leads, and write the reasoning lead by lead. Finish by naming the rhythm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C05-BD1

Build the AV block table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C05 Activity: NPLEX-P1-CARD-07-01-C05-BD1 Date: _____

On blank paper build a table of the four AV blocks with one row each for first degree, Mobitz I, Mobitz II, and third degree. Use the columns PR interval, site of block, QRS width, and risk of progression, and fill every cell. Beside the Mobitz I row, sketch three beats and the dropped beat and write what happens to the RR intervals before the drop and why. Beside the third degree row, write what the P waves and QRS complexes are doing relative to each other and what that pattern is called. Finish by listing the causes of AV block from the notes, including the infectious one and the antibody that injures the fetal AV node.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C05-BD2

Gradual failure against all-or-none failure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C05 Activity: NPLEX-P1-CARD-07-01-C05-BD2 Date: _____

Write the ion channel that carries the upstroke in AV nodal cells and the one that carries it in His-Purkinje cells, then explain why one tissue fails gradually and the other fails suddenly. Use that difference to explain, in two or three sentences each: why the PR lengthens beat to beat in Mobitz I, why P waves drop without warning in Mobitz II, and why the escape rhythm after infranodal block is slower and less reliable than the escape after nodal block. Then a patient has 2:1 block with a QRS of 0.14 s. Write why the PR interval cannot settle the site of block here and which feature points to where it is. Finish by writing what a bifascicular block becomes when the last fascicle fails, with the rate and QRS width you expect.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-07-01-C05-CP01

Right against left bundle branch block

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	RBBB	LBBB
Direction of septal depolarization		
QRS shape in V1		
QRS shape in I and V6		
Septal q wave in I, aVL, V5, V6		
Splitting of the second heart sound		
QRS duration in complete block and in incomplete block		
Usual causes and what the finding implies about the heart		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C06-BD1

Mechanisms of tachyarrhythmia, and the premature beats

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C06 Activity: NPLEX-P1-CARD-07-01-C06-BD1 Date: _____

Write the three mechanisms that generate tachyarrhythmias. Under each one, write what is happening at the membrane and give the triggers listed in the notes. For reentry, write the three conditions a circuit needs and the formula for wavelength, then write which two changes shorten the wavelength and make reentry easier to sustain. Next, draw a premature atrial contraction and a premature ventricular contraction as they look on a strip. Beside each one write the P wave, the QRS width, the T wave direction, and whether the following pause is compensatory, and explain each pause from what the premature impulse does to the SA node.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C06-BD2

Read four strips back to their mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C06 Activity: NPLEX-P1-CARD-07-01-C06-BD2 Date: _____

Four patients present with a fast rhythm. Patient 1 has an irregularly irregular narrow QRS at 130 bpm with no visible P waves. Patient 2 has a regular narrow QRS at 148 bpm with sawtooth atrial waves at about 300 per minute. Patient 3 has a regular narrow QRS at 190 bpm that began and stopped abruptly, with no P wave visible before the QRS. Patient 4 has a regular wide QRS at 170 bpm with P waves marching through it independently. For each patient name the rhythm and its mechanism, write where in the heart the circuit or focus sits, and explain from the atrial rate and nodal refractoriness why the ventricular rate is what it is. For patient 1, write what is lost from ventricular filling and where stasis forms. For patient 4, write the other two findings that confirm the origin.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-07-01-C06-RT01

Rebuild the reentrant circuit

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C06 Activity: NPLEX-P1-CARD-07-01-C06-RT01 Date: _____

Draw a branch point with two limbs and rebuild the reentrant circuit in six steps, numbering each step on the drawing and writing it out in words. Mark on the drawing where the unidirectional block sits and where conduction must be slow. Then write the two changes that stop the circuit. Underneath, redraw the circuit as the AV node in nodal reentry: label the fast pathway and the slow pathway with their refractory periods, show which one the premature beat goes down and which one it comes back up, and write where the retrograde P wave ends up on the tracing.

Rebuild from memory: A reentrant circuit

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-07-01-C06-PR01

Predict the effect of a low plasma potassium

A woman taking a diuretic has a plasma K+ of 2.6 mEq/L (reference range 3.5 to 5.0 mEq/L) and is started on a macrolide antibiotic. Predict each item below, write the direction of the change, and give the current or channel that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the repolarizing K+ current IKr		
What happens to the duration of the ventricular action potential and to the QT interval		
What happens to the U wave		
What happens to L-type Ca2+ channels late in the prolonged plateau		
What happens to the rhythm once those reopened channels drive a depolarization to threshold		
What the resulting rhythm looks like on the tracing and what it can degenerate into		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-07-01-C07-BD1

Preexcitation and the failing sinus node

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C07 Activity: NPLEX-P1-CARD-07-01-C07-BD1 Date: _____

Draw a heart in outline with the AV node and an accessory pathway crossing the fibrous ring. Write the embryological reason the pathway exists and the tissue it is made of, then rebuild in six steps how an impulse crossing it produces the ECG pattern. Draw the resulting complex and label the short PR interval with its value, the delta wave, and the wide fusion QRS, and write which part of the complex each pathway produced. Separately, write what sick sinus syndrome is, its usual cause, and the two findings that follow from having fewer pacemaker cells.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-07-01-C07-BD2

The dangerous pathway and the long pause

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-7 Conduction System and the Electrocardiogram
Competency: NPLEX-P1-CARD-07-01-C07 Activity: NPLEX-P1-CARD-07-01-C07-BD2 Date: _____

An accessory pathway with a refractory period of about 0.25 s carries atrial fibrillation to the ventricles. Write how many impulses per minute it can pass, why the AV node would not pass that many, and what the tracing looks like in rate, regularity, QRS width, and QRS shape. Write why this rhythm can turn into ventricular fibrillation, and why slowing the AV node does not slow the ventricular rate. Then compare orthodromic and antidromic reentry using the pathway: for each, write the direction of travel, the QRS width, and whether the delta wave is visible. Finally, two patients each have a pause after a run of atrial fibrillation. One pause is an exact multiple of the P to P interval and one is not. Write what each pause tells you about where the sinus node failed, and explain from overdrive suppression why the pause follows the tachycardia.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-07-01-C07-CA01

Palpitations in a young man with a wide, irregular tachycardia

A 24-year-old man is brought to the emergency department because of 40 minutes of palpitations and lightheadedness that began during a soccer match. He has had brief episodes of rapid regular palpitations since adolescence, each starting and stopping abruptly. Temperature is 37.0 °C (98.6 °F), pulse is 230/min and irregular, respirations are 22/min, and blood pressure is 88/54 mm Hg. The electrocardiogram shows a wide-complex tachycardia that is irregularly irregular, with QRS complexes that vary in width and shape from beat to beat and no identifiable P waves. An electrocardiogram taken two years earlier, during sinus rhythm, showed a PR interval of 0.10 s, a slurred upstroke at the start of a QRS of 0.13 s, and ST-T changes directed opposite to the QRS.

1. Recognize. Name the atrial rhythm and the pathway carrying it to the ventricles, and name the finding on the earlier electrocardiogram that identifies that pathway.

2. Explain. Explain why the ventricular rate is this fast and why the QRS complexes vary in width and shape from beat to beat.

3. Differentiate. Write how this tracing differs from atrial fibrillation conducted through the AV node and from ventricular tachycardia, naming the features on the strip that separate them.

4. Investigate. Write which findings on the earlier sinus rhythm electrocardiogram you would measure to confirm the pathway, and what value or appearance each one would have.

5. Decide. State which property of the accessory pathway makes this rhythm dangerous, and what this rhythm can turn into.

6. Justify. Justify that answer from the refractory periods of the accessory pathway and the AV node, and explain why slowing conduction through the AV node would not reduce the ventricular rate.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-07-01-C03-Q01

Complete block after an inferior infarction

A 68-year-old man comes to the emergency department because of 3 hours of chest pain. Pulse is 42/min and blood pressure is 86/58 mm Hg. The electrocardiogram shows ST elevation in leads II, III, and aVF, P waves at a rate of 84/min, and QRS complexes at a rate of 44/min with a duration of 0.08 s, unrelated to the P waves. Which of the following best explains the bradycardia in this patient?

A. Necrosis of the His bundle and the proximal bundle branches

B. Ischemia of the atrioventricular node with adenosine release and reflex vagal discharge

C. Conduction of atrial impulses through an accessory pathway

D. Degenerative fibrosis of the sinoatrial node with overdrive suppression

Why did you pick it?

Board-style question NPLEX-P1-CARD-07-01-C05-Q02

Locating the level of block

A 72-year-old woman comes to the physician because of two episodes of syncope. The electrocardiogram shows a PR interval of 0.18 s that is identical in every conducted beat, with occasional P waves that are followed by no QRS complex. The conducted QRS complexes have a duration of 0.14 s with an rSR prime pattern in lead V1. Which of the following is the most likely site of the conduction failure?

A. Sinoatrial node

B. Atrial muscle

C. Atrioventricular node

D. His bundle and bundle branches

Why did you pick it?